# MSG No. 75.6 $P_I4$ [ Type IV, tetragonal ]

Table 1: Wyckoff site: `2a`, site symmetry: `4..`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\, 0,\, z]$ | `[1,2,3,4]` |
| 2 | $\left[\frac{1}{2},\, \frac{1}{2},\, z+\frac{1}{2}\right]$ | `[5,6,7,8]` |

Table 2: Wyckoff site: `4b`, site symmetry: `2..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\, \frac{1}{2},\, z\right]$ | `[1,4]` |
| 2 | $\left[\frac{1}{2},\, 0,\, z\right]$ | `[2,3]` |
| 3 | $\left[\frac{1}{2},\, 0,\, z+\frac{1}{2}\right]$ | `[5,8]` |
| 4 | $\left[0,\, \frac{1}{2},\, z+\frac{1}{2}\right]$ | `[6,7]` |

Table 3: Wyckoff site: `8c`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\, y,\, z]$ | `[1]` |
| 2 | $[-y,\, x,\, z]$ | `[2]` |
| 3 | $[y,\, -x,\, z]$ | `[3]` |
| 4 | $[-x,\, -y,\, z]$ | `[4]` |
| 5 | $\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$ | `[5]` |
| 6 | $\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$ | `[6]` |
| 7 | $\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$ | `[7]` |
| 8 | $\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$ | `[8]` |